therefore reduced to 0.5). Second, memorize the following conversions between decibels and *amplitude* ratios:

60dB	=	1000
40dB	=	100
20dB	=	10
0dB	=	1
-20dB	=	0.1
-40dB	=	0.01
-60dB	=	0.001

How Information is Represented in Signals

The most important part of any DSP task is understanding how *information* is contained in the signals you are working with. There are many ways that information can be contained in a signal. This is especially true if the signal is manmade. For instance, consider all of the modulation schemes that have been devised: AM, FM, single-sideband, pulse-code modulation, pulse-width modulation, etc. The list goes on and on. Fortunately, there are only two ways that are common for information to be represented in naturally occurring signals. We will call these: **information represented in the time domain**, and **information represented in the frequency domain**.

Information represented in the time domain describes when something occurs and what the amplitude of the occurrence is. For example, imagine an experiment to study the light output from the sun. The light output is measured and recorded once each second. Each sample in the signal indicates what is happening at that instant, and the level of the event. If a solar flare occurs, the signal directly provides information on the time it occurred, the duration, the development over time, etc. Each sample contains information that is interpretable without reference to any other sample. Even if you have only one sample from this signal, you still know something about what you are measuring. This is the simplest way for information to be contained in a signal.

In contrast, information represented in the frequency domain is more indirect. Many things in our universe show periodic motion. For example, a wine glass struck with a fingernail will vibrate, producing a ringing sound; the pendulum of a grandfather clock swings back and forth; stars and planets rotate on their axis and revolve around each other, and so forth. By measuring the frequency, phase, and amplitude of this periodic motion, information can often be obtained about the system producing the motion. Suppose we sample the sound produced by the ringing wine glass. The fundamental frequency and harmonics of the periodic vibration relate to the mass and elasticity of the material. A single sample, in itself, contains no information about the periodic motion, and therefore no information about the wine glass. The information is contained in the *relationship* between many points in the signal.